

KISANKAVACH

Karnataka

Software Requirements Specification

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Farmer Credit. Benefits. Protection.

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Note: Page references are represented by section numbering to preserve stability across approved revisions.

1. Purpose and Scope

Defines the Karnataka pilot baseline for farmer-centred Kisan Credit Card enablement, benefit visibility, consent-driven data exchange, grievance handling, fraud protection and programme oversight.

In scope: farmer mobile experience, assisted service, participating-bank workflow, Government command centre, authorised integrations and operational reporting.

2. Guiding Principles

Kannada-first and low-literacy friendly; reuse verified information; explicit purpose-bound consent; least privilege; interoperability through adapters; configurable workflows and auditability.

The bank remains the final credit decision-maker. Government-originated and bank-originated data retain their source and provenance.

3. System Context



External authorities remain systems of record. KisanKavach stores references, snapshots and provenance needed for the authorised purpose.

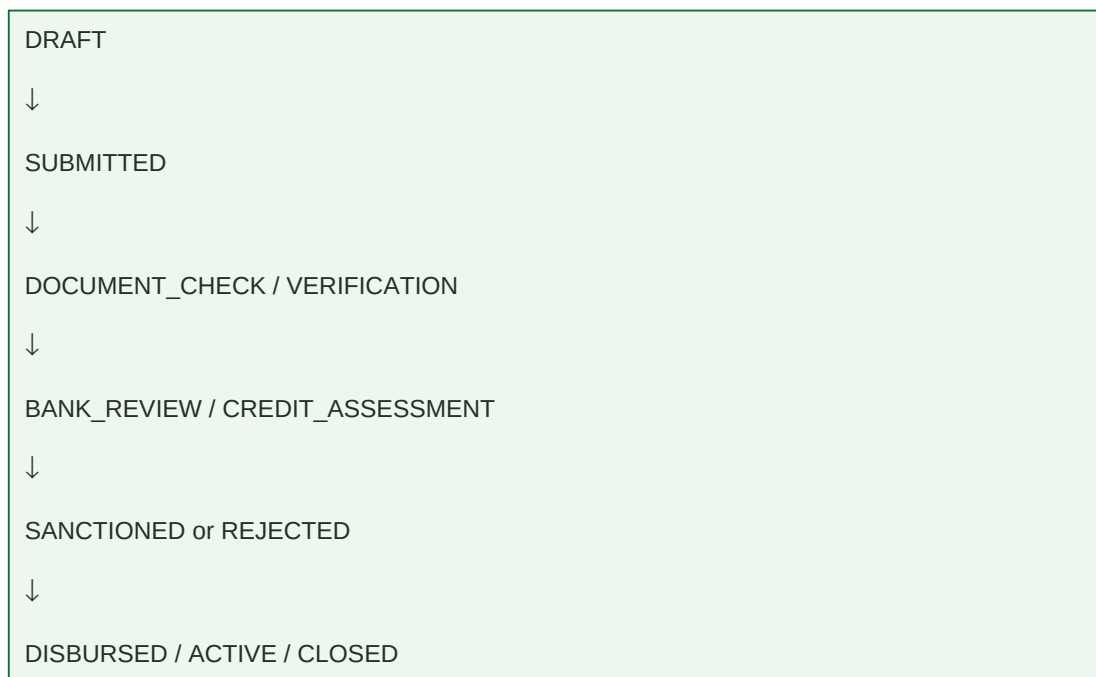
4. Actors and Access

Actor	Permitted scope	Strong controls
Farmer	Own profile, applications, benefits, consents and grievances	OTP, session and device risk
Bank officer	Assigned/branch applications	MFA, role and maker-checker where configured
Government officer	Authorised geography and aggregation level	MFA, scoped RBAC and export control
Administrator	Configuration and operations	Privileged access management and audit
Auditor	Read-only approved evidence	Time-bound access

5. Functional System Requirements

ID	The system shall
FR-001	authenticate supported users and enforce role/scope authorization
FR-002	capture, version, validate and withdraw purpose-specific consent
FR-003	retrieve and label authorised farmer/land/crop data with provenance
FR-004	save application drafts and validate each wizard step
FR-005	submit idempotently and allocate a unique application identifier
FR-006	route applications to the selected bank/branch and start stage timers
FR-007	support document upload, quarantine, malware scan and access control
FR-008	record requests, decisions, reasons, sanction and disbursement events
FR-009	present plain-language timeline and action-required messages
FR-010	create, assign, escalate and close grievances
FR-011	send multilingual, preference-aware notifications
FR-012	produce scoped operational and analytical reports

6. KCC State Model



Every transition requires an allowed source state, authorised actor, timestamp, reason where applicable, correlation ID and immutable audit entry. Repeated inbound events must not create duplicate transitions.

7. Validation and Error Behaviour

Validate at client and server; server rules are authoritative. Errors use stable codes, a farmer-safe message, correlation ID and retry guidance. External timeouts must not lose a draft or create duplicate submissions.

Class	Behaviour
Validation	Highlight field; preserve entries
Authentication	Generic response; throttle and audit
Dependency unavailable	Retry safely; show temporary-unavailability message

Class	Behaviour
Conflict	Return latest state and reconciliation instruction
Security rejection	Block, audit and avoid sensitive detail

8. Non-Functional Requirements

Performance: p95 core reads under 2 seconds and writes under 3 seconds excluding partner delay. Scalability: horizontal stateless services and queued workloads. Resilience: multi-AZ production, backups, tested recovery. Observability: metrics, structured logs, traces and business events without exposed secrets.

9. Data, Privacy and Security

Data minimisation, purpose limitation, encryption, key rotation, secure secrets, masked non-production data, configurable retention, consent evidence, export controls, tamper-resistant audit and incident response are mandatory.

10. Interfaces and Notifications

Support versioned REST/JSON APIs, signed webhooks or controlled batch modes. Kannada and English message templates must identify the application, status and safe next step without exposing sensitive financial or identity data.

11. Acceptance Criteria

Requirements are accepted only when positive, negative, permission, accessibility, resilience and audit tests pass; critical defects are closed; operational dashboards and alerts are demonstrated; and traceability links requirements to test evidence.

End of controlled baseline